SPACE X | APPLIED DATA SCIENCE CAPSTONE

Falcon 9 First-Stage Landing Prediction

- Prepared by: Premanand (edit if needed before submission).
- Objective: analyze historical Falcon 9 launches and build a classification model to predict first-stage landing outcome.
- Project code repository: https://github.com/premanand404/spacex-data-science-capstone
- Data source: IBM Skills Network course teaching dataset.

Repository: https://github.com/premanand404/spacex-data-science-capstone

Executive Summary

Business question and analytical approach

- Reusable first stages are a key contributor to launch-cost reduction. Estimating landing success can inform mission planning and competitive launch bids.
- This project combines data collection, cleaning, exploratory analysis, SQL queries, geographic visualization, an interactive dashboard, and classification.
- Model comparison is implemented for Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.
- Important: run the included code and replace any unreported metrics with measured results before final submission.

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Introduction

Why predict landing outcomes?

- Falcon 9 missions differ by launch site, payload, orbit, booster configuration, and mission date.
- Target variable: Class (1 = successful first-stage landing; 0 = unsuccessful) in the course dataset.
- Goal: identify useful associations and evaluate whether available pre-launch features support prediction.
- Scope is limited to the course teaching dataset and should not be interpreted as a complete operational SpaceX risk model.

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Data Collection Methodology

Reproducible dataset acquisition

- Use the IBM Skills Network dataset URL documented in the repository code.
- The project data is loaded into Pandas; dates are parsed and the Year field is derived for trend analysis.
- The course dataset is a curated instructional dataset rather than a live production feed.
- Data collection and web-scraping notebook sections should be run and documented with the actual API/Wikipedia output if your course rubric specifically requires them.

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Data Wrangling Methodology

Prepare consistent analytical fields

- Inspect data types, dimensions, missing values, and duplicate rows.
- Convert date and payload fields to appropriate types; derive year from launch date.
- Preserve the binary target Class and document the target mapping.
- For modelling, impute missing numeric values, one-hot encode categorical predictors, and scale features for distance/margin-based models.
- Exclude target and post-outcome fields such as Outcome/LandingPad to reduce leakage.

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EDA & Visualization Methodology

Questions answered by plots

- Compare launch counts and landing success rates across launch sites.
- Plot yearly landing success rate to explore changes over time.
- Compare payload mass distributions between successful and unsuccessful outcomes.
- Always interpret rates alongside group counts; very small groups can produce unstable percentages.

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EDA Results — Launch Sites

Run notebook 01 to populate the exact values

- Grouped analysis calculates launch count and mean Class for each launch site.
- Chart output: outputs/success_by_site.png (generated by src/analyze_spacex.py).
- Interpret site differences with caution: launch sites may serve different mission types and payload/orbit mixes.
- Insert the generated chart and state the actual highest/lowest observed rates after running the code.

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EDA Results — Yearly Trend & Payload

Run notebook 01 / analysis script

- Yearly trend uses the mean of Class for each launch year.
- Payload comparison uses boxplots split by Class to show spread, median, and outliers.
- Chart outputs: outputs/success_by_year.png and outputs/payload_by_class.png.
- Use the measured chart patterns; do not infer causality from descriptive comparisons.

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EDA with SQL

SQLite queries included in the project

- Query 1: SELECT LaunchSite, COUNT(*) AS launches, AVG(Class) AS success_rate FROM launches GROUP BY LaunchSite.
- Query 2: SELECT Orbit, COUNT(*) AS launches, AVG(Class) AS success_rate FROM launches GROUP BY Orbit HAVING COUNT(*) >= 3.
- Query 3: group by launch year to compare volume and success rate.
- SQL outputs are written to outputs/sql_results.txt when src/analyze_spacex.py runs.

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Folium Launch Map

Interactive geographic exploration

- src/folium_map.py plots launch coordinates from the course dataset.
- Markers are colored by observed outcome and show flight number, launch site, and status in the popup.
- Output: outputs/launch_map.html. Open it in a browser after running the script.
- Geographic proximity analysis can be added by calculating distances from launch pads to nearby coastline/roads, but distance claims should be supported by coordinate sources.

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Plotly Dash Interactive Analytics

Dashboard for exploratory filtering

- src/dashboard.py provides a launch-site dropdown and payload-mass range slider.
- Visuals include a landing-outcome pie chart and payload-mass vs flight-number scatter plot.
- Run python src/dashboard.py; open the local address printed in the terminal.
- Dashboard filters are exploratory and do not imply that payload alone causes landing success.

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Predictive Analysis — Classification

Train/test split and four models

- Target: Class. Use stratified 80/20 train/test split with fixed random seed 42.
- Models: Logistic Regression, Support Vector Machine, Decision Tree, and K-Nearest Neighbors.
- Scaling is applied in pipelines for Logistic Regression, SVM, and KNN.
- Evaluation includes accuracy, precision, recall, F1 score, and confusion matrix.

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Predictive Results & Confusion Matrix

Measured results must be generated locally

- Run python src/analyze_spacex.py or execute notebook 02_predictive_analysis.ipynb.
- outputs/model_metrics.csv contains measured test metrics; outputs/model_comparison.png visualizes them.
- outputs/confusion_matrix.csv and outputs/classification_report.txt show best-by-F1 model results.
- Name the best model only after reviewing actual metrics; accuracy alone can hide class imbalance.

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Conclusion

What this workflow delivers

- A reproducible pipeline for launch-site, yearly, payload, SQL, geographic, dashboard, and classification analysis.
- Success rates can vary by mission context; group sizes and mission mix must be considered before operational conclusions.
- Model selection should weigh precision and recall, not accuracy alone, and results require validation beyond a single train/test split.
- Future work: cross-validation, hyperparameter tuning, more recent verified launch data, and calibration of predicted probabilities.

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Creativity & Innovative Insights

Translate analysis into useful decisions

- Use the dashboard to compare site and payload subsets interactively.
- Use year and orbit breakdowns to investigate whether performance changes align with changes in booster reuse or mission profile.
- Prioritize interpretability and uncertainty: show group counts beside success rates and report confusion-matrix errors.
- Add a cost-impact scenario only when cost assumptions are sourced and clearly labelled.

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Repository & Reproducibility

Code, notebooks, and generated outputs

- Repository: https://github.com/premanand404/spacex-data-science-capstone
- Files to upload: README.md, requirements.txt, src/, notebooks/, data/ (course datasets only if permitted), and generated outputs needed to support conclusions.
- Run the code, inspect all generated charts/metrics, then update this presentation with the exact results.
- Only submit this PDF after running and verifying outputs. A scaffold without measured results may not satisfy the rubric.

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